VOLTARAX
PRIVATE LIMITED
POWERING A SUSTAINABLE FUTURE

COMPANY BROCHURE · 2026

Making solar smarter with AI and bio-coated intelligence.

Voltarax builds VoltVision AI, a platform that reads the surface health of solar panels, predicts coating degradation and optimises clean-energy output, using low-cost sensing and edge AI.

- DPIIT-recognised startup
- Udyam-registered
- Renewable solar energy

WHO WE ARE

Deeptech for smarter, longer-lasting solar

Voltarax Private Limited is a Delhi-registered deeptech startup (incorporated 23 August 2026) working where solar energy, bio-based coatings and artificial intelligence meet. VoltVision AI pairs electrical and 18-channel spectral sensing with machine learning, so operators see what is happening on the panel surface **before** the loss shows up in the energy bill.

4	AI models in one platform
18	spectral channels, 410–940 nm
30 s	sensor sampling interval
₹13,730	pilot hardware bill of materials

COMPANY AT A GLANCE

INCORPORATED	23 August 2026, Private Limited Company
STARTUP RECOGNITION	DPIIT certificate no. DIPP285459
MSME REGISTRATION	Udyam UDYAM-DL-06-0228493, Micro enterprise
SECTOR	Renewable solar energy; electronic components (NIC 26109)
REGISTERED OFFICE	Shastri Nagar, North West Delhi, 110052
PILOT SITE	Planned 32-panel array, Jaipur, Rajasthan
FLAGSHIP PRODUCT	VoltVision AI v2.1 (live demo at voltarax.onrender.com)

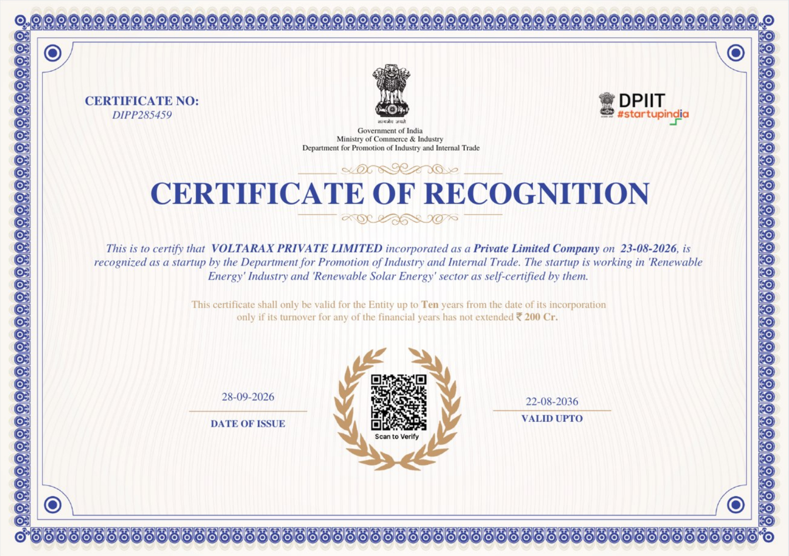
COMPANY REGISTRATIONS & GOVERNMENT RECOGNITION

Verified credentials

Voltarax Private Limited is a formally incorporated, DPIIT-recognised startup developing sustainable energy solutions through bio-based materials, solar technology and artificial intelligence. The registrations below support its legal identity and startup status for incubation, pilot deployment and commercialisation.

DPIIT Startup Recognition

Startup India · DPIIT



CERTIFICATE NO:
DIPP285459

सत्यमेव जयते
Government of India
Ministry of Commerce & Industry
Department for Promotion of Industry and Internal Trade

DPIIT
#startupindia

CERTIFICATE OF RECOGNITION

*This is to certify that **VOLTARAX PRIVATE LIMITED** incorporated as a **Private Limited Company** on **23-08-2026**, is recognized as a startup by the Department for Promotion of Industry and Internal Trade. The startup is working in 'Renewable Energy' Industry and 'Renewable Solar Energy' sector as self-certified by them.*

This certificate shall only be valid for the Entity up to **Ten** years from the date of its incorporation only if its turnover for any of the financial years has not extended **₹ 200 Cr.**

28-09-2026
DATE OF ISSUE

Scan to Verify

22-08-2036
VALID UPTO

CERTIFICATE NO.
DIPP285459

INCORPORATED
23 Aug 2026

RECOGNITION ISSUED
28 Sep 2026

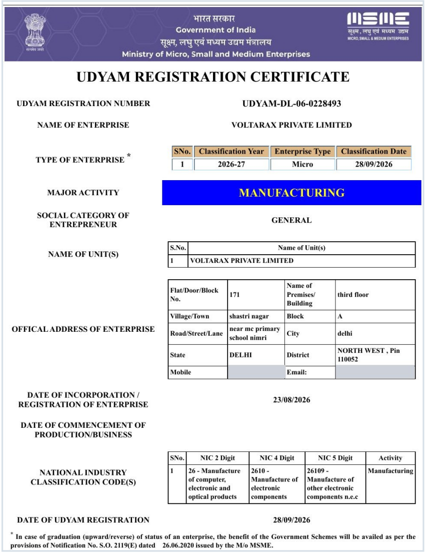
VALID UP TO
22 Aug 2036

SECTOR (SELF-CERTIFIED)
Renewable Solar Energy

Recognised by the Department for Promotion of Industry and Internal Trade, Government of India, under the Startup India initiative. The certificate carries a scan-to-verify QR code.

MSME / Udyam Registration

Government of India



भारत सरकार
Government of India
सूक्ष्म, लघु एवं मध्यम उद्यम मंत्रालय
Ministry of Micro, Small and Medium Enterprises

MSME
सूक्ष्म, लघु एवं मध्यम उद्यम
MICRO, SMALL & MEDIUM ENTERPRISES

UDYAM REGISTRATION CERTIFICATE

UDYAM REGISTRATION NUMBER	UDYAM-DL-06-0228493
NAME OF ENTERPRISE	VOLTARAX PRIVATE LIMITED

TYPE OF ENTERPRISE *

SNo.	Classification Year	Enterprise Type	Classification Date
1	2026-27	Micro	28/09/2026

MAJOR ACTIVITY	MANUFACTURING
SOCIAL CATEGORY OF ENTREPRENEUR	GENERAL

NAME OF UNIT(S)

S.No.	Name of Unit(s)
1	VOLTARAX PRIVATE LIMITED

OFFICAL ADDRESS OF ENTERPRISE

Flat/Door/Block No.	171	Name of Premises/ Building	third floor
Village/Town	shastri nagar	Block	A
Road/Street/Lane	near mc primary school nimri	City	delhi
State	DELHI	District	NORTH WEST , Pin 110052
Mobile	[redacted]	Email:	[redacted]

DATE OF INCORPORATION / REGISTRATION OF ENTERPRISE	23/08/2026
DATE OF COMMENCEMENT OF PRODUCTION/BUSINESS	

NATIONAL INDUSTRY CLASSIFICATION CODE(S)

SNo.	NIC 2 Digit	NIC 4 Digit	NIC 5 Digit	Activity
1	26 - Manufacture of computer, electronic and optical products	2610 - Manufacture of electronic components	26109 - Manufacture of other electronic components n.e.c	Manufacturing

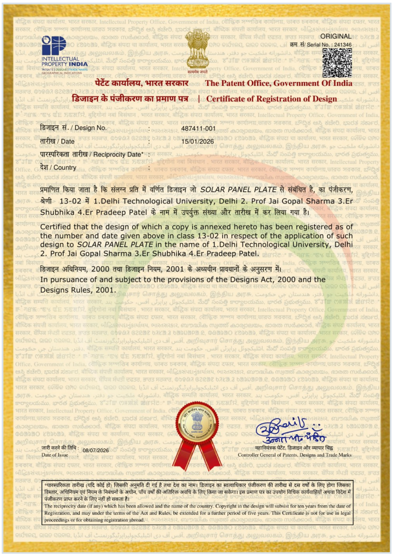
DATE OF UDYAM REGISTRATION	28/09/2026

* In case of graduation (upward/reverse) of status of an enterprise, the benefit of the Government Schemes will be availed as per the provisions of Notification No. S.O. 2119(E) dated 26.06.2020 issued by the M/o MSME.

Registered under the Udyam framework for Micro, Small and Medium Enterprises as a Micro enterprise in manufacturing (NIC 26109). Udyam no. UDYAM-DL-06-0228493. Contact details are redacted in this copy.

Design Registration

Designs Act, 2000



INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

सत्यमेव जयते

ORIGINAL
क्रम सं/ Serial No. : 241346

पेटेंट कार्यालय, भारत सरकार The Patent Office, Government Of India
डिजाइन के पंजीकरण का प्रमाण पत्र | Certificate of Registration of Design

डिजाइन सं. / Design No. : 487411-001
तारीख / Date : 15/01/2026
पारस्परिकता तारीख / Reciprocity Date* :
देश / Country :

प्रमाणित किया जाता है कि संलग्न प्रति में वर्णित डिजाइन जो *SOLAR PANEL PLATE* से संबंधित है, का पंजीकरण, श्रेणी 13-02 में 1.Delhi Technological University, Delhi 2. Prof Jai Gopal Sharma 3.Er Shubhika 4.Er Pradeep Patel के नाम में उपर्युक्त संख्या और तारीख में कर लिया गया है।

Certified that the design of which a copy is annexed hereto has been registered as of the number and date given above in class 13-02 in respect of the application of such design to *SOLAR PANEL PLATE* in the name of 1.Delhi Technological University, Delhi 2. Prof Jai Gopal Sharma 3.Er Shubhika 4.Er Pradeep Patel.

डिजाइन अधिनियम, 2000 तथा डिजाइन नियम, 2001 के अध्यधीन प्रावधानों के अनुसरण में।
In pursuance of and subject to the provisions of the Designs Act, 2000 and the Designs Rules, 2001.

जारी करने की तिथि : 08/07/2026
Date of Issue :

महानियंत्रक पेटेंट, डिजाइन और व्यापार चिह्न
Controller General of Patents, Designs and Trade Marks

*पारस्परिकता तारीख (यदि कोई हो) जिसकी अनुमति दी गई है तथा देश का नाम। डिजाइन का स्वत्वाधिकार पंजीकरण की तारीख से दस वर्षों के लिए होगा जिसका विस्तार, अधिनियम एवं नियम के निबंधनों के अधीन, पाँच वर्षों की अतिरिक्त अवधि के लिए किया जा सकेगा। इस प्रमाण पत्र का उपयोग विधिक कार्यवाहियों अथवा विदेश में पंजीकरण प्राप्त करने के लिए नहीं हो सकता है।
The reciprocity date (if any) which has been allowed and the name of the country. Copyright in the design will subsist for ten years from the date of Registration, and may under the terms of the Act and Rules, be extended for a further period of five years. This Certificate is not for use in legal proceedings or for obtaining registration abroad.

Design No. 487411-001, class 13-02, "Solar Panel Plate", registered 15 Jan 2026 in the names of Delhi Technological University, Prof. Jai Gopal Sharma, Er. Shubhika and Er. Pradeep Patel.

✓ **Due diligence ready.** Original certificates are available on request. All three documents are issued by Government of India authorities.

THE CHALLENGE

Solar panels lose output quietly

Surface losses

Dust, micro-cracks and coating wear reduce yield, yet most monitoring only sees electrical output.

Late detection

By the time the power curve drops, the cause is already weeks old, so cleaning and recoating are reactive.

Coating blind spot

Protective coatings degrade under UV, and few operators can measure how much coating life is left.

OUR APPROACH

VoltVision AI watches the panel surface, not just the power meter

01 BFCI: Bio-Fluorescent Coating Intelligence

18-channel NIR/UV/visible sensing (AS7265x) feeds a Bayesian-optimised bagging model with SHAP explainability. Outputs a coating health score, UV absorption and days to recoat.

Score 0–100% · Recoat alert below 75%

02 Dual-domain surface inspection

A camera classifier labels each panel clean, dusty, cracked or bio-degraded every 30 minutes, with an 85% confidence gate before auto-alerts. Built for edge deployment on Raspberry Pi.

4 surface classes · Edge-ready

03 Energy forecasting

Predicts generation at 1 hour, 6 hours, 24 hours and 7 days from irradiance, temperature, humidity, air quality and battery state, refreshed every 60 seconds.

4 horizons · 60 s refresh

04 Two-layer anomaly detection

Z-score rules on each ESP32 node (30 s) plus an Isolation Forest on the gateway (60 s) flag voltage drops, over-temperature, open circuits and shading.

Node + gateway · 4 fault types

HOW IT FITS TOGETHER

SENSE
ESP32 nodes, AS7265x, INA226, DHT22, BH1750

→

EDGE AI
Raspberry Pi 4 gateway running the four models

→

STREAM
MQTT messaging and InfluxDB time-series

→

SERVE
FastAPI with JWT login and role-based access

→

ACT
Live dashboard, alerts, work orders, relay control

Bio-based coating

Panels are treated with a natural blend of mango (60%) and pomegranate (40%) extract. BFCI tells operators when that coating needs renewing, targeting a 75–90 day recoat cycle.

Carbon and ESG reporting

The dashboard converts generation into CO_2 offset using the India CEA grid factor of 0.82 kg/kWh, with monthly charts ready for ESG and sustainability reports.

THE TEAM

Three co-founders, from the lab to the field

SH Er. Shubhika

Co-Founder & Research Lead

Delhi Technological University

Leads research direction and technical validation of the BFCI and inspection models.

DK Dinesh Kumar

Co-Founder, CTO & CSO

Dronacharya Group of Institutions

Leads technology vision and platform engineering across AI, backend and dashboard.

PP Er. Pradeep Patel

Co-Founder & Hardware Lead

Delhi Technological University

Leads sensing hardware, firmware and field deployment of the pilot array.

> **Academic collaboration:** Prof. Jai Gopal Sharma, Delhi Technological University, is a co-registrant of Design No. 487411-001 (Solar Panel Plate) together with DTU and two of our co-founders.

WHERE WE ARE

A working prototype, with field validation next

Built and running

- ✓ FastAPI platform with REST API, JWT login and role-based access
- ✓ 4 AI models trained and running inside the platform
- ✓ 11-page operations dashboard: panels, alerts, forecast, relay, carbon
- ✓ ESP32 sensor firmware and AS7265x spectral reader written
- ✓ 68 of 68 automated tests passing; live demo deployed

Next, with funding

- Install sensors on the 32-panel Jaipur array
- Collect 60–90 days of real data and retrain models
- Train MobileNetV3 on 500+ real images per class
- Reinforcement-learning energy dispatch using coating health
- 30-day validation and a research paper submission
- Extend IP protection before public demos

Transparency note: *the models are currently trained on physics-based simulated Jaipur solar data. Real-sensor validation is the milestone this round is designed to reach.*

ROADMAP

1 MONTHS 1–2

Hardware pilot

Order and wire sensor nodes, flash firmware, bring up the gateway and stream live data.

2 MONTHS 3–4

Real-data retraining

Retrain forecasting after 60 days and BFCI after 90 days; collect panel images.

3 MONTH 5

Validation and IP

Run a 30-day validation (MAPE, F1, R^2), submit a paper, extend patent filings.

4 BEYOND

Scale

Multi-site rollout and bio-state reinforcement-learning energy management.

RESEARCH AGENDA

BFCI on real data

Spectral AI for bio-coating degradation in PV panels, with MAPE and R^2 on field data.

Dual-domain CV

One CNN with UV and visible inputs, benchmarked against two separate models.

Bio-state RL-EMS

A PPO agent with coating health in its state space for energy dispatch.

Carbon impact

Quantified CO_2 offset gain from AI-optimised BioSolar versus a baseline.

WHY VOLTARAX

Why this is worth a conversation

1 Measures what others miss

BFCI reads coating health directly through spectral sensing, instead of inferring problems from lost output after the fact.

2 Capital-light to pilot

A full sensing and gateway bill of materials of ₹13,730 keeps the first field pilot small and fast.

3 Full stack already built

Firmware, edge AI, API, dashboard and alerts exist today, so new funding goes to validation rather than rebuilding.

4 Credible foundations

DPIIT-recognised and Udyam-registered, with a registered design and research ties to Delhi Technological University.

PARTNER WITH US

Help us take VoltVision AI from simulation to the field

Angel investors

Seed funding for the hardware pilot, real-data validation and IP extension. Round details are shared directly with interested investors.

Pilot clients

Solar plant, rooftop and EPC partners who can host a pilot and give operating feedback on a live site.

DTU alumni network

Mentorship, introductions to customers and advice on scaling a deeptech company from campus to market.

NEXT STEPS

1. **Intro call** Share goals and fit →
2. **Live demo** Walk through VoltVision AI →
3. **Diligence** Originals and technical deep-dive →
4. **Pilot or investment** Agree the way forward

GET IN TOUCH

Web	voltarax.onrender.com
Email	shubhikabt@gmail.com
Registered office	171, 3rd Floor, Block A, Shastri Nagar, Delhi 110052
Pilot site	Jaipur, Rajasthan

Academic ecosystem

DTU Innovation and Incubation Foundation

POWERING A SUSTAINABLE FUTURE